

Africast-Time Series Analysis & Forecasting Using R

<https://workshop.f4sg.org/africast/>

5. Forecasting with regression, how to
represent temporal structure with regressors



Outline

- 1 The linear model with time series
- 2 Evaluating the regression model
- 3 Selecting predictors
- 4 Forecasting with regression
- 5 Correlation, causation and forecasting
- 6 Some useful predictors for regression models

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Regression models

- To **explain**
- To **forecast**

- Simple linear regression model(SLR)
- Multiple linear regression model (MLR)

SLR model in theory

Regression model allows for a linear relationship between the forecast variable y and a single predictor variable x .

$$y_t = \beta_0 + \beta_1 x_t + \varepsilon_t.$$

- y_t is the variable we want to predict: the response variable
- Each x_t is numerical and is called a predictor
- β_0 and β_1 are regression coefficients

SLR model in practice

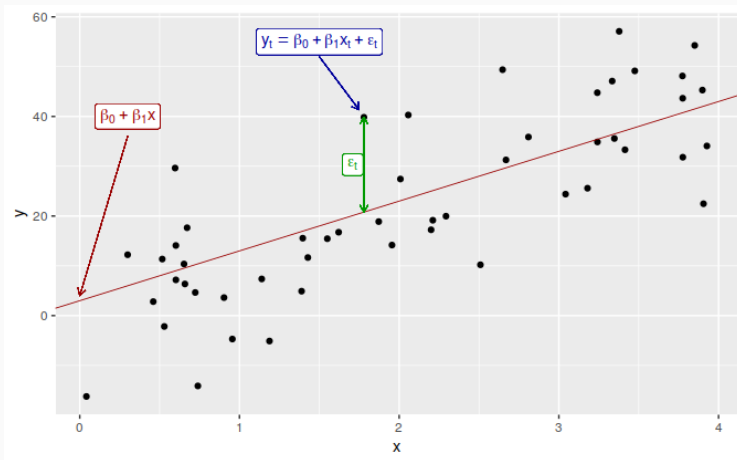
In practice, of course, we have a collection of observations but we do not know the values of the coefficients $\hat{\beta}_0, \hat{\beta}_1$. These need to be estimated from the data.

$$y_t = \hat{\beta}_0 + \hat{\beta}_1 x_t.$$

- y_t is the response variable
- Each x_t is a predictor
- $\hat{\beta}_0$ is the estimated intercept
- $\hat{\beta}_1$ is the estimated slope

What is the best fit

- There are many ways that a straight line can be laid on the scatter
- Best known criterion is called Ordinary Least Squares(OLS)



Estimation of the model

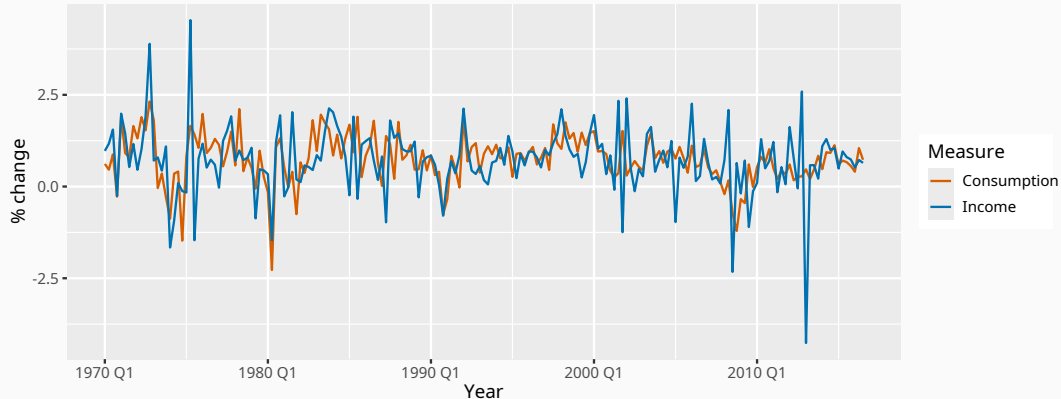
That is, we find the values of β_0 and β_1 which minimise

$$\sum_{i=1}^N e_i^2 = \sum_{i=1}^N (y_i - \beta_0 - \beta_1 x_i)^2.$$

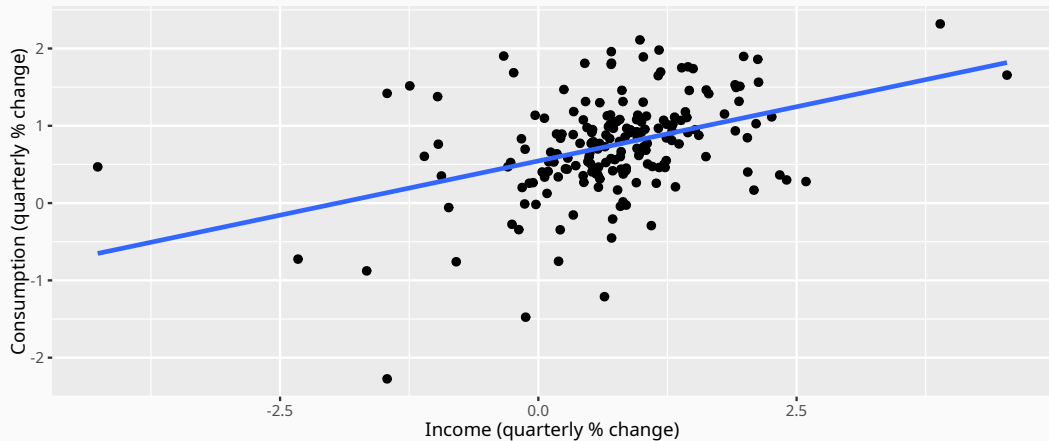
- This is called *least squares* estimation because it gives the least value of the sum of squared errors.
- Finding the best estimates of the coefficients is often called *fitting* the model to the data.
- We refer to the *estimated* coefficients using the notation $\hat{\beta}_0, \hat{\beta}_1$.

Example: US consumption expenditure

```
us_change %>%  
  gather("Measure", "Change", Consumption, Income) %>%  
  autoplot(Change) +  
  ylab("% change") + xlab("Year")
```



Example: US consumption expenditure



Example: US consumption expenditure

```
fit_cons <- us_change %>%  
  model(lm = TSLM(Consumption ~ Income))  
report(fit_cons)
```

Series: Consumption

Model: TSLM

Residuals:

Min	1Q	Median	3Q	Max
-2.408	-0.318	0.026	0.300	1.452

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.5451	0.0557	9.79	< 2e-16 ***
Income	0.2806	0.0474	5.91	1.6e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.603 on 185 degrees of freedom

Multiple R-squared: 0.159, Adjusted R-squared: 0.154

F-statistic: 35 on 1 and 185 DF, p-value: 2e-08

Multiple regression

- In multiple regression there is one variable to be forecast and several predictor variables.
- The basic concept is that we forecast the time series of interest y assuming that it has a linear relationship with other time series $x_1, x_2, \dots, x_K$
- We might forecast daily A&E attendance y using temperature x_1 and GP visits x_2 as predictors.

How many variable can we add?

You can add as many as you want but be aware of:

- Over-fitting
- Multicollinearity

Multiple regression and forecasting

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \varepsilon_t.$$

- y_t is the variable we want to predict: the response variable
- Each $x_{j,t}$ is numerical and is called a predictor. They are usually assumed to be known for all past and future times.
- The coefficients $\beta_1, \dots, \beta_k$ measure the effect of each predictor after taking account of the effect of all other predictors in the model.

That is, the coefficients measure the **marginal effects**.

- ε_t is a white noise error term

Estimation of the model

We find the values of $\hat{\beta}_0, \dots, \hat{\beta}_k$ which minimise

$$\sum_{i=1}^N e_i^2 = \sum_{i=1}^N (y_i - \beta_0 - \beta_1 x_{1,i} - \dots - \beta_k x_{k,i})^2.$$

- This is called *least squares* estimation because it gives the least value of the sum of squared errors
- Finding the best estimates of the coefficients is often called *fitting* the model to the data
- We refer to the *estimated* coefficients using the notation $\hat{\beta}_0, \dots, \hat{\beta}_k$.

Useful predictors in linear regression

Linear trend

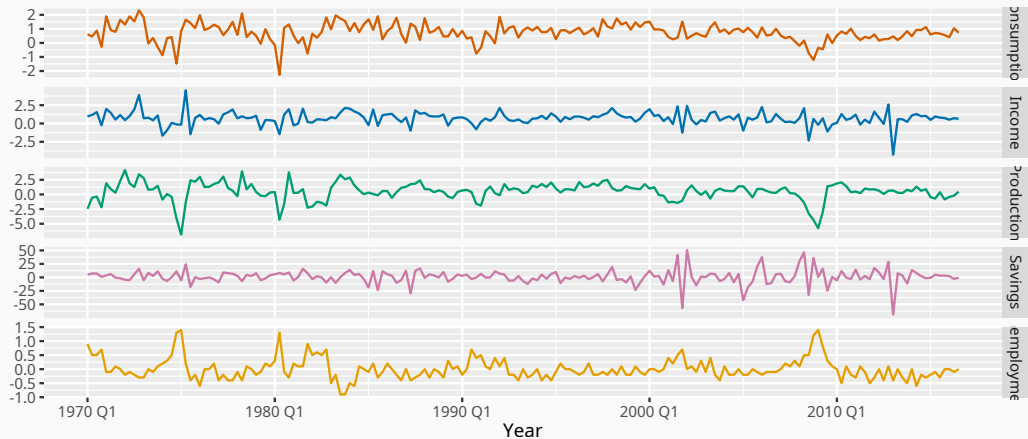
$$x_t = t$$

- $t = 1, 2, \dots, T$
- Strong assumption that trend will continue.
- use special function `trend()`

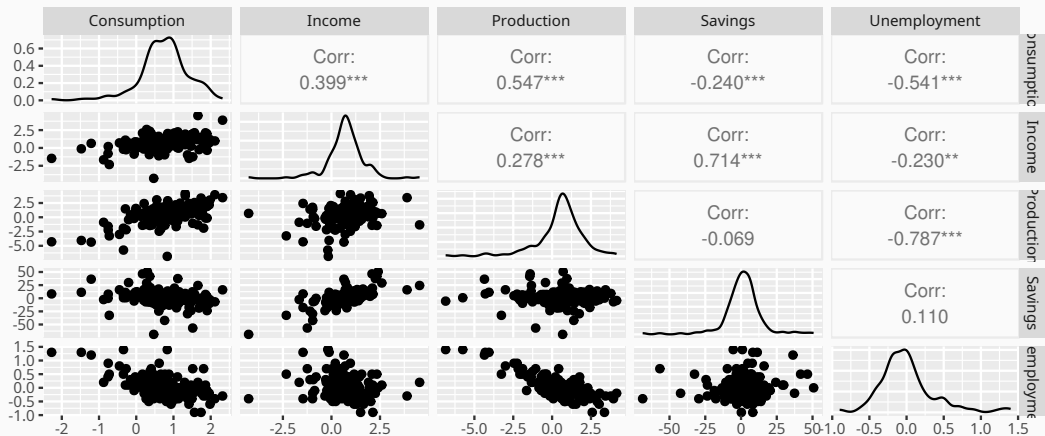
Seasonality

- Seasonality will be considered based on the interval of index
- use special function `season()`

Example: US consumption expenditure



Example: US consumption expenditure



Example: US consumption expenditure

```
fit_consMR <- us_change %>%  
  model(lm = TSLM(Consumption ~ Income + Production + Unemployment + Savings))  
report(fit_consMR)
```

Series: Consumption

Model: TSLM

Residuals:

Min	1Q	Median	3Q	Max
-0.883	-0.176	-0.037	0.153	1.206

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.26729	0.03721	7.18	1.7e-11	***
Income	0.71448	0.04219	16.93	< 2e-16	***
Production	0.04589	0.02588	1.77	0.078	.
Unemployment	-0.20477	0.10550	-1.94	0.054	.
Savings	-0.04527	0.00278	-16.29	< 2e-16	***

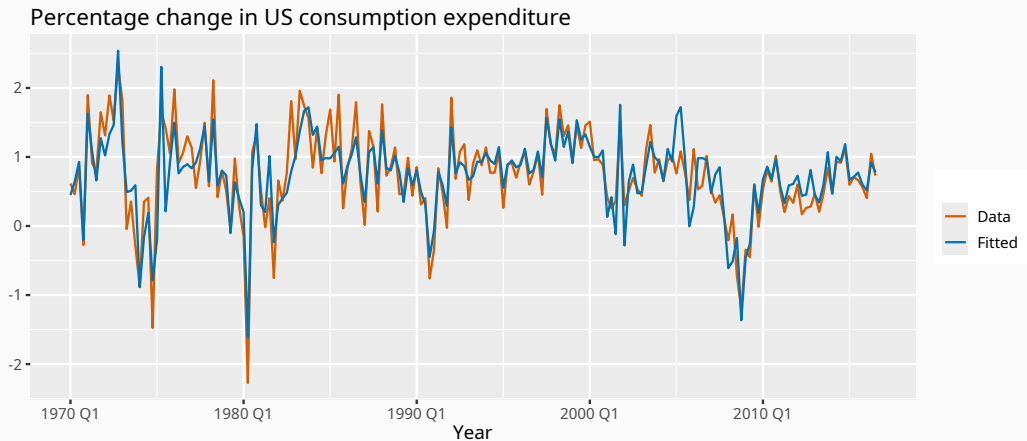
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.329 on 182 degrees of freedom

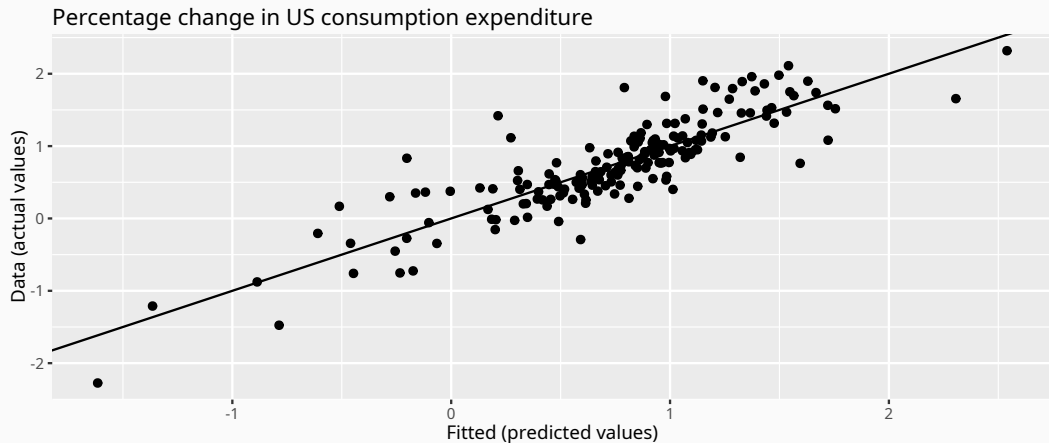
Multiple R-squared: 0.754, Adjusted R-squared: 0.749

F-statistic: 139 on 4 and 182 DF, p-value: <2e-16

Example: US consumption expenditure

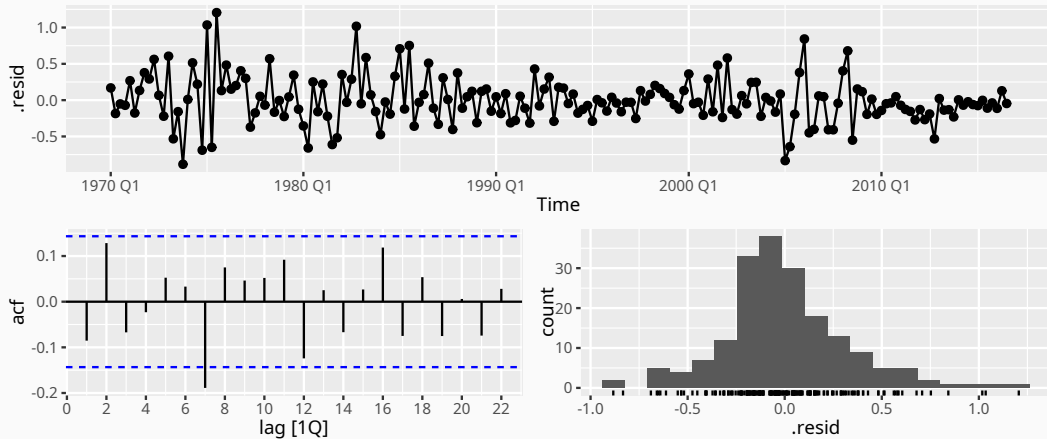


Example: US consumption expenditure



Example: US consumption expenditure

```
augment(fit_consMR) %>%  
  gg_tsdisplay(.resid, plot_type="hist")
```



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Multiple regression and forecasting

For forecasting purposes, we require the following assumptions:

- ε_t are uncorrelated and zero mean
- ε_t are uncorrelated with each $x_{j,t}$.

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It is **useful** to also have $\varepsilon_t \sim N(0, \sigma^2)$ when producing prediction intervals or doing statistical tests.

Residual diagnostics

There are a series of plots that should be produced in order to check different aspects of the fitted model and the underlying assumptions.

- 1 check if residuals are uncorrelated using ACF
- 2 Check if residuals are normally distributed

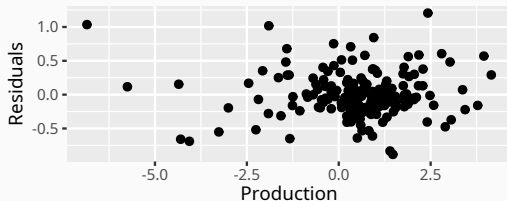
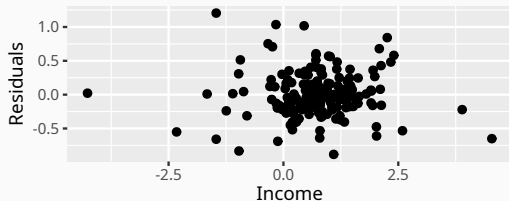
Residual scatterplots

Useful for spotting outliers and whether the linear model was appropriate.

- Scatterplot of residuals ε_t against each predictor $x_{j,t}$.
- Scatterplot residuals against the fitted values $\hat{y}_t$
- Expect to see scatterplots resembling a horizontal band with no values too far from the band and no patterns such as curvature or increasing spread.

Example: US consumption expenditure

```
df <- left_join(us_change, residuals(fit_consMR), by = "Time")
p1 <- ggplot(df, aes(x=Income, y=.resid)) +
  geom_point() + ylab("Residuals")
p2 <- ggplot(df, aes(x=Production, y=.resid)) +
  geom_point() + ylab("Residuals")
p3 <- ggplot(df, aes(x=Savings, y=.resid)) +
  geom_point() + ylab("Residuals")
p4 <- ggplot(df, aes(x=Unemployment, y=.resid)) +
  geom_point() + ylab("Residuals")
(p1 | p2) / (p3 | p4)
```



Residual patterns

- If a plot of the residuals vs any predictor in the model shows a pattern, then the relationship is non-linear.
- If a plot of the residuals vs any predictor **not** in the model shows a pattern, then the predictor should be added to the model.
- If a plot of the residuals vs fitted values shows a pattern, then there is heteroscedasticity in the errors. (Could try a transformation.)

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Comparing regression models

Computer output for regression will always give the R^2 value. This is a useful summary of the model.

- It is equal to the square of the correlation between y and $\hat{y}$.
- It is often called the “coefficient of determination”.
- It can also be calculated as follows:
$$R^2 = \frac{\sum(\hat{y}_t - \bar{y})^2}{\sum(y_t - \bar{y})^2}$$
- It is the proportion of variance accounted for (explained) by the predictors.

Comparing regression models

However ...

- R^2 does not allow for degrees of freedom.
- Adding *any* variable tends to increase the value of R^2 , even if that variable is irrelevant.

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To overcome this problem, we can use *adjusted* R^2 :

$$\bar{R}^2 = 1 - (1 - R^2) \frac{T - 1}{T - k - 1}$$

where k = no. predictors and T = no. observations.

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where k = no. predictors and T = no. observations.

Maximising $\bar{R}^2$ is equivalent to minimising $\hat{\sigma}^2$.

$$\hat{\sigma}^2 = \frac{1}{T - k - 1} \sum_{t=1}^T \epsilon_t^2$$

Cross-validation

- 1 Remove observation t from the data set, and fit the model using the remaining data. Then compute the error for the omitted observation
- 2 Repeat step 1 for $t = 1, \dots, T$
- 3 Compute the MSE from errors obtained in 1. We shall call this the CV

Akaike's Information Criterion

$$\text{AIC} = -2 \log(L) + 2(k + 2)$$

where L is the likelihood and k is the number of predictors in the model.

- This is a *penalised likelihood* approach.
- *Minimising* the AIC gives the best model for prediction.
- AIC penalises terms more heavily than $\bar{R}^2$.
- Minimising the AIC is asymptotically equivalent to minimising MSE via leave-one-out cross-validation.

Corrected AIC

For small values of T , the AIC tends to select too many predictors, and so a bias-corrected version of the AIC has been developed.

$$\text{AIC}_c = \text{AIC} + \frac{2(k+2)(k+3)}{T-k-3}$$

As with the AIC, the AIC_c should be minimised.

Comparing regression models

```
glance(fit_consMR) %>%  
  select(r_squared, adj_r_squared, AIC, AICc, CV)
```

```
# A tibble: 1 x 5
```

	r_squared	adj_r_squared	AIC	AICc	CV
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0.754	0.749	-409.	-409.	0.116

Choosing regression variables

Best subsets regression

- Fit all possible regression models using one or more of the predictors.
- Choose the best model based on one of the measures of predictive ability (CV, AIC, AICc).

Choosing regression variables

Backwards stepwise regression

- Start with a model containing all variables.
- Try subtracting one variable at a time. Keep the model if it has lower CV or AICc.
- Iterate until no further improvement.
- You can also do forward stepwise

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Ex-ante versus ex-post forecasts

- *Ex ante forecasts* are made using only information available in advance.
 - ▶ require forecasts of predictors
- *Ex post forecasts* are made using later information on the predictors.
 - ▶ useful for studying behaviour of forecasting models.
- trend, seasonal and calendar variables are all known in advance, so these don't need to be forecast.

Scenario based forecasting

- Assumes possible scenarios for the predictor variables
- Prediction intervals for scenario based forecasts do not include the uncertainty associated with the future values of the predictor variables.

Building a predictive regression model

- If getting forecasts of predictors is difficult, you can use lagged predictors instead.

$$\beta_0 + \beta_1 x_{1,t-h} + \cdots + \beta_k x_{k,t-h} + \varepsilon_t.$$

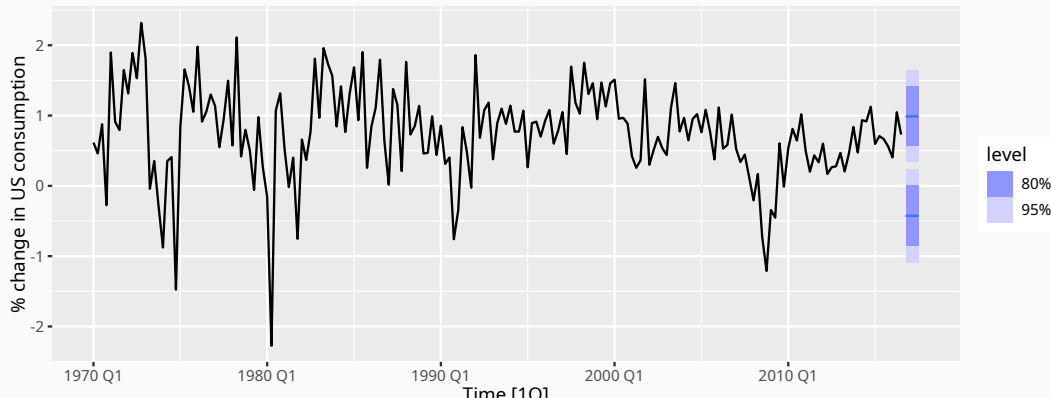
- A different model for each forecast horizon h .

US Consumption

```
fit_consBest <- us_change %>%  
  model(  
    TSLM(Consumption ~ Income + Savings + Unemployment)  
  )  
  
down_future <- new_data(us_change, 4) %>%  
  mutate(Income = -1, Savings = -0.5, Unemployment = 0)  
fc_down <- forecast(fit_consBest, new_data = down_future)  
  
up_future <- new_data(us_change, 4) %>%  
  mutate(Income = 1, Savings = 0.5, Unemployment = 0)  
fc_up <- forecast(fit_consBest, new_data = up_future)
```

US Consumption

```
us_change %>% autoplot(Consumption) +  
  ylab("% change in US consumption") +  
  autolayer(fc_up, series = "increase") +  
  autolayer(fc_down, series = "decrease") +  
  guides(colour = guide_legend(title = "Scenario"))
```

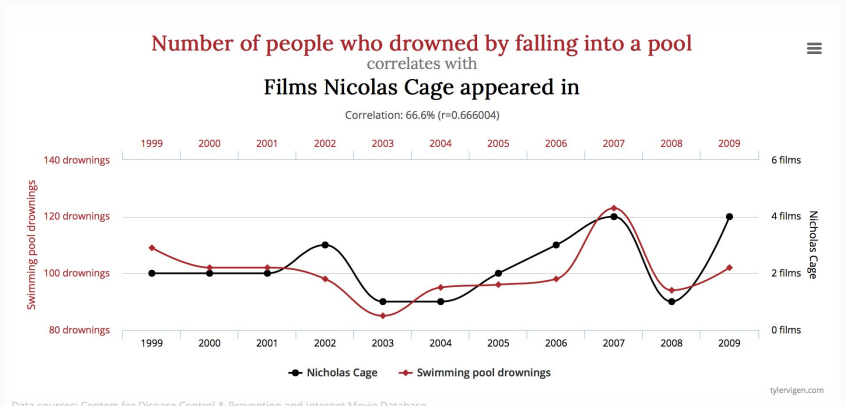


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Correlation does not imply causation

Check out <https://www.tylervigen.com/spurious-correlations>



Correlation is not causation

- When x is useful for predicting y , it is not necessarily causing y .
- e.g., predict number of drownings y using number of ice-creams sold x .
- Correlations are useful for forecasting, even when there is no causality.
- Better models usually involve causal relationships (e.g., temperature x and people z to predict drownings y).

Multicollinearity

In regression analysis, multicollinearity occurs when:

- Two predictors are highly correlated (i.e., the correlation between them is close to ± 1).
- A linear combination of some of the predictors is highly correlated with another predictor.
- A linear combination of one subset of predictors is highly correlated with a linear combination of another subset of predictors.

Multicollinearity

If multicollinearity exists...

- the numerical estimates of coefficients may be wrong (worse in Excel than in a statistics package)
- don't rely on the p -values to determine significance.
- there is no problem with model *predictions* provided the predictors used for forecasting are within the range used for fitting.
- omitting variables can help.
- combining variables can help.

Outliers and influential observations

Things to watch for

- *Outliers*: observations that produce large residuals.
- *Influential observations*: removing them would markedly change the coefficients. (Often outliers in the x variable).
- *Lurking variable*: a predictor not included in the regression but which has an important effect on the response.
- Points should not normally be removed without a good explanation of why they are different.

Modern regression models

- Suppose instead of 3 regressors we had 44.
 - ▶ For example, 44 predictors leads to 18 trillion possible models!
- Stepwise regression cannot solve this problem due to the number of variables.
- We need to use the family of Lasso models: lasso, ridge, elastic net

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Dummy variables

If a categorical variable takes only two values (e.g., 'Yes' or 'No'), then an equivalent numerical variable can be constructed taking value 1 if yes and 0 if no. This is called a **dummy variable**.

	A	B
1	Yes	1
2	Yes	1
3	No	0
4	Yes	1
5	No	0
6	No	0
7	Yes	1
8	Yes	1
9	No	0
10	No	0
11	No	0
12	No	0
13	Yes	1
14	No	0

Dummy variables

If there are more than two categories, then the variable can be coded using several dummy variables (one fewer than the total number of categories).

	A	B	C	D	E
1	Monday	1	0	0	0
2	Tuesday	0	1	0	0
3	Wednesday	0	0	1	0
4	Thursday	0	0	0	1
5	Friday	0	0	0	0
6	Monday	1	0	0	0
7	Tuesday	0	1	0	0
8	Wednesday	0	0	1	0
9	Thursday	0	0	0	1
10	Friday	0	0	0	0
11	Monday	1	0	0	0
12	Tuesday	0	1	0	0
13	Wednesday	0	0	1	0
14	Thursday	0	0	0	1
15	Friday	0	0	0	0

Beware of the dummy variable trap!

- Using one dummy for each category gives too many dummy variables!
- The regression will then be singular and inestimable.
- Either omit the constant, or omit the dummy for one category.
- The coefficients of the dummies are relative to the omitted category.

Uses of dummy variables

Seasonal dummies

- For quarterly data: use 3 dummies
- For monthly data: use 11 dummies
- For daily data: use 6 dummies

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Outliers

- If there is an outlier, you can use a dummy variable to remove its effect.

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Outliers

- If there is an outlier, you can use a dummy variable to remove its effect.

Public holidays

- For daily data: if it is a public holiday, $\text{dummy}=1$, otherwise $\text{dummy}=0$.

Intervention variables

Spikes

- Equivalent to a dummy variable for handling an outlier.

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Steps

- Variable takes value 0 before the intervention and 1 afterwards.

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Change of slope

- Variables take values 0 before the intervention and values $\{1, 2, 3, \dots\}$ afterwards.
- this could be also handled using `trend()`

Include special event using dummies

- Christmas Eve: if Christmas Eve, $v_t = 1$, $v_t = 0$ otherwise
- New year's Day: if New year's Day, $v_t = 1$, $v_t = 0$ otherwise.
- and more: Ramadan and Chinese new year, school holiday, etc

Interactions

For example, sometimes the effect of a particular event might be different if it is on a weekend or a week day or its effect might be different in each shift:

- you need to introduce an interaction variable
- you can use a new dummy as : $v1*v2$

Lagged predictors

The model include present and past values of predictor:

$$x_t, x_{t-1}, x_{t-2}, \dots$$

$$y_t = a + \beta_0 x_t + \beta_1 x_{t-1} + \dots + \beta_k x_{t-k} + \varepsilon_t$$

- x can influence y , but y is not allowed to influence x .

Lagged predictors

- Lagged values of a predictor:
 - ▶ Create new variables by shifting the existing variable backwards

Example: x is advertising which has a delayed effect

x_1 = advertising for previous month;

x_2 = advertising for two months previously;

$\vdots$

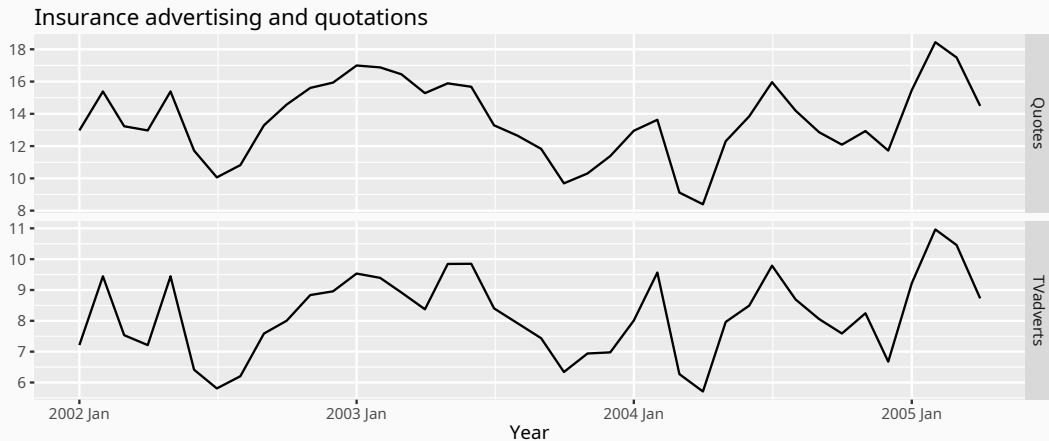
x_m = advertising for m months previously.

Example: Insurance quotes and TV adverts

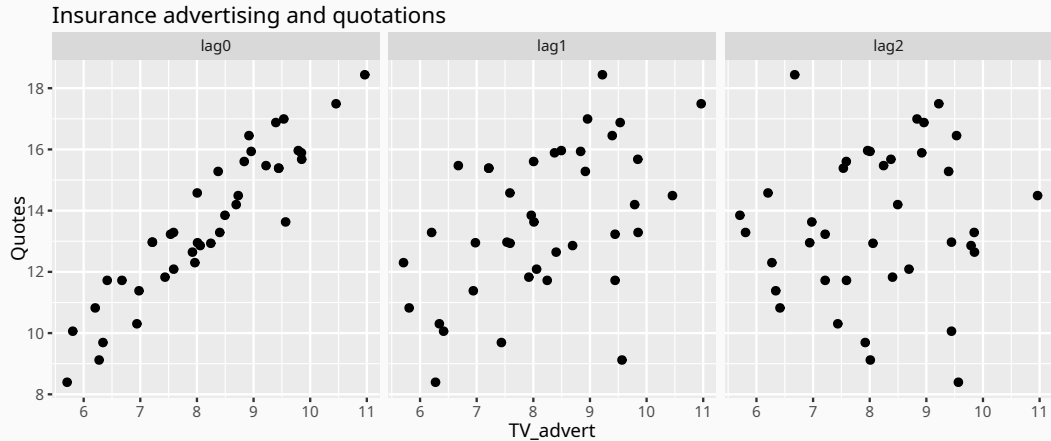
```
insurance
```

```
# A tsibble: 40 x 3 [1M]
      Month Quotes TVadverts
    <mth>   <dbl>    <dbl>
1 2002 Jan   13.0     7.21
2 2002 Feb   15.4     9.44
3 2002 Mar   13.2     7.53
4 2002 Apr   13.0     7.21
5 2002 May   15.4     9.44
6 2002 Jun   11.7     6.42
7 2002 Jul   10.1     5.81
8 2002 Aug   10.8     6.20
9 2002 Sep   13.3     7.59
10 2002 Oct   14.6     8.00
# i 30 more rows
```

Example: Insurance quotes and TV adverts



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Example: Insurance quotes and TV adverts

```
fit <- insurance |>
  # Restrict data so models use same fitting period
  mutate(Quotes = c(NA, NA, NA, Quotes[4:40])) |>
  model(
    ARIMA(Quotes ~ pdq(d = 0) + TVadverts),
    ARIMA(Quotes ~ pdq(d = 0) + TVadverts +
      lag(TVadverts)),
    ARIMA(Quotes ~ pdq(d = 0) + TVadverts +
      lag(TVadverts) +
      lag(TVadverts, 2))),
    ARIMA(Quotes ~ pdq(d = 0) + TVadverts +
      lag(TVadverts) +
      lag(TVadverts, 2) +
      lag(TVadverts, 3))
  )
```

Example: Insurance quotes and TV adverts

```
glance(fit)
```

Lag order	sigma2	log_lik	AIC	AICc	BIC
0	0.265	-28.3	66.6	68.3	75.0
1	0.209	-24.0	58.1	59.9	66.5
2	0.215	-24.0	60.0	62.6	70.2
3	0.206	-22.2	60.3	65.0	73.8

Example: Insurance quotes and TV adverts

```
# Re-fit to all data
fit <- insurance |>
  model(ARIMA(Quotes ~ TVadverts + lag(TVadverts) + pdq(d = 0)))
report(fit)
```

Series: Quotes

Model: LM w/ ARIMA(1,0,2) errors

Coefficients:

	ar1	ma1	ma2	TVadverts	lag(TVadverts)	intercept
	0.512	0.917	0.459	1.2527	0.1464	2.16
s.e.	0.185	0.205	0.190	0.0588	0.0531	0.86

sigma^2 estimated as 0.2166: log likelihood=-23.9

AIC=61.9 AICc=65.4 BIC=73.7

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	0.512	0.917	0.459	1.2527	0.1464	2.16
s.e.	0.185	0.205	0.190	0.0588	0.0531	0.86

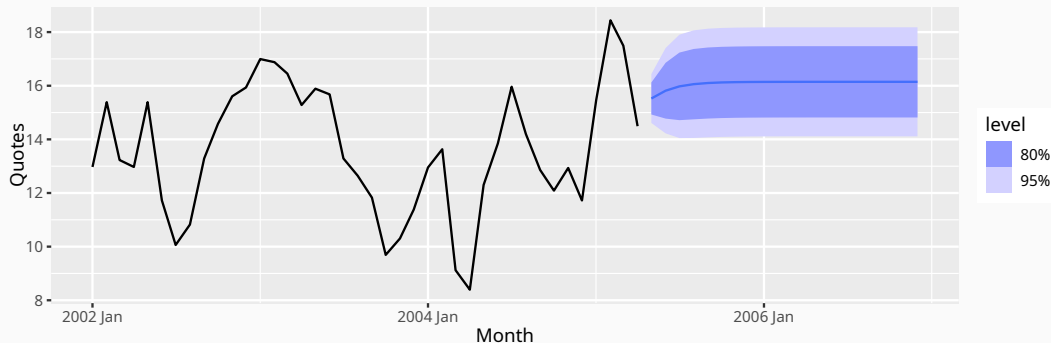
sigma^2 estimated as 0.2166: log likelihood=-23.9

AIC=61.9 AICc=65.4 BIC=73.7

$$y_t = 2.16 + 1.25x_t + 0.15x_{t-1} + \eta_t,$$
$$\eta_t = 0.512\eta_{t-1} + \varepsilon_t + 0.92\varepsilon_{t-1} + 0.46\varepsilon_{t-2}.$$

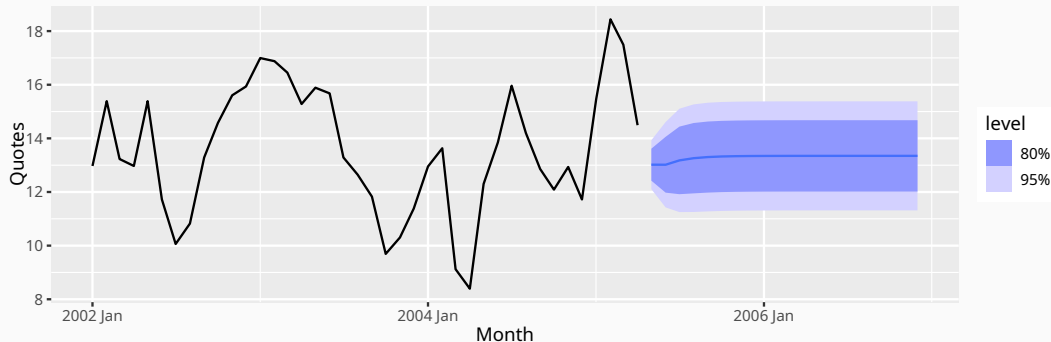
Example: Insurance quotes and TV adverts

```
advert_a <- new_data(insurance, 20) |>  
  mutate(TVadverts = 10)  
forecast(fit, advert_a) |> autoplot(insurance)
```



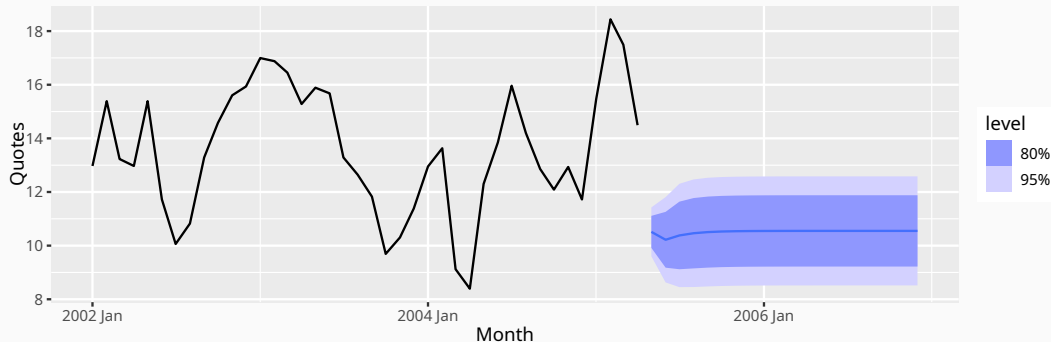
Example: Insurance quotes and TV adverts

```
advert_b <- new_data(insurance, 20) |>  
  mutate(TVadverts = 8)  
forecast(fit, advert_b) |> autoplot(insurance)
```



Example: Insurance quotes and TV adverts

```
advert_c <- new_data(insurance, 20) |>  
  mutate(TVadverts = 6)  
forecast(fit, advert_c) |> autoplot(insurance)
```



Lead predictors

Sometimes a change in the predictor x_t that will happen in the future will affect the value of y_t in the past. We say x_t is a leading indicator.

- Lead values of a predictor:

- ▶ Create new variables by shifting the existing variable forwards

- $y_t = \text{sales}$, $x_t = \text{tax policy announcement}$